BUILDING YOUR FIRST ML PROJECT

From a real-world question → to a working prediction

RUNNING EXAMPLE · PREDICT A STUDENT'S FINAL EXAM MARKS

Problem → Data → Model → Evaluation → Deployment

So... where do we actually start?

Not with Linear Regression. Not with a fancy model. Start with the question.

THE REAL TASK

Turn a real-world question

Example → “Can we predict a student’s final exam marks?”

TODAY’S ROUTE

01	**QUESTION**	What are we predicting?
02	**DATA**	What do we have?
03	**EVIDENCE**	What is messy / useful?
04	**MODEL**	What fits the task?
05	**EVALUATE**	Does it generalize?
06	**IMPROVE**	What should change next?

The model is one decision inside a much bigger reasoning process.

1 · Problem formulation

Before code: define the prediction.

“Can we predict a student's final exam marks?”

INPUT FEATURES X

hours_studied
hours_slept
attendance
previous_marks

MODEL

learns a mapping

$y = f(X)$

TARGET y

final_marks

continuous number

REGRESSION because the output is a continuous numerical value.

2 · Understand the dataset

EDA is where surprises become visible.

SHAPE
rows / columns

DTYPES
numeric? text? dates?

RANGES
are values plausible?

MISSING
where are NaNs?

DUPS
duplicates?

REAL DATA IS MESSY

"six" instead of 6
inspect → understand → fix

NaN values
inspect → understand → fix

850% attendance
inspect → understand → fix

Male / M / MALE
inspect → understand → fix

3 · Features + target

Decide what the model is allowed to see.

FEATURES X
hours_studied
hours_slept
attendance
prev_marks

These are the inputs available before prediction.

TARGET y

final_marks

This is the ground-truth outcome we want to predict.

Never leak the target into X.

X = INPUTS

y = OUTPUT

4 · Train / test split

Generalize. Don't memorize.

TRAINING DATA

Used to learn the model's parameters.

The model is allowed to see these rows during training.

TEST DATA

Held back until evaluation.

It represents unseen data and checks generalization.

TRAIN **TEST**

Tune the model on validation data — keep test data untouched.

5 · Regression or classification?

The target determines the task type.

REGRESSION

Predict a continuous number.

Example:
Hours studied + attendance + previous marks
→ final marks = 78.5

Typical example model: Linear Regression

CLASSIFICATION

Predict a category / label.

Example:
CGPA + projects + internships + coding score
→ placement = YES / NO

Typical example model: Logistic Regression

6 · Training → the model learns

Fit parameters by reducing prediction error.

X + y

training examples

TRAINING ALGORITHM

predict → measure error →
adjust parameters → repeat

FITTED MODEL

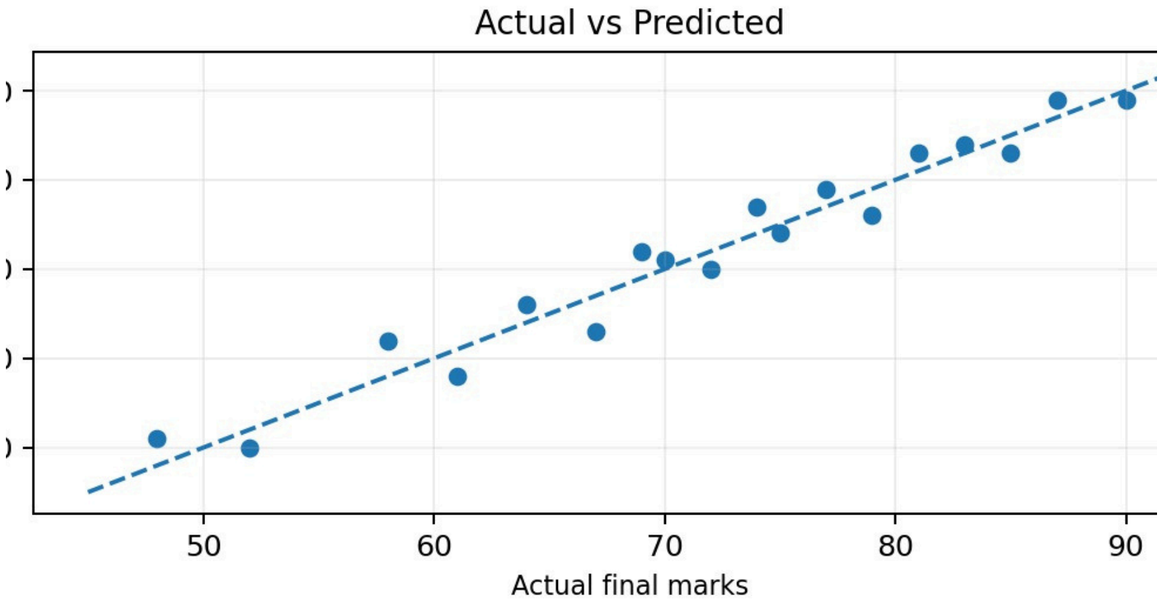
learned weights / coefficients

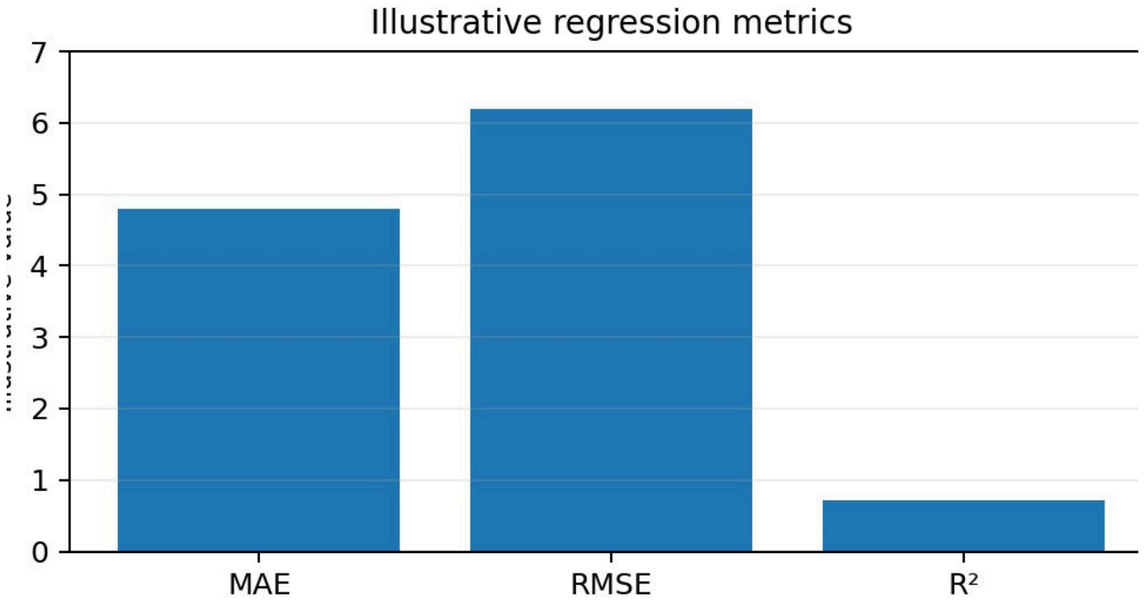
Each parameter controls how much a feature influences the prediction.

LOSS / ERROR

7 · Evaluation: how wrong are we?

Different tasks need different metrics.





REGRESSION

MAE → average absolute error|RMSE → larger errors matter more | R^2 → explained variation

Accuracy can fool you

A model can look “95% correct”while being useless for the thing you care about.

WHEN ACCURACY LOOKS GREAT...



“95% accuracy?!”

THE DUMB MODEL

Predict “NOT PLACED” for everyone.

100 students → only 5 are placed

95 correct • 5 missed

95% ACCURACY

0% RECALL

The model never finds a placed student. → Always check the metric against the real goal.

Overfitting

Train 99%→Test 58%: The model memorizes instead of generalizing.



TRAIN
99% — extremely comfortable with data it has already seen.

TEST
58% — unseen data exposes weak generalization.

MEMORIZE ≠ LEARN

Final challenge · Will a student get placed?

Apply the same thinking to a new problem.

student_id	**CGPA**	**Attendance**	**Num_Projects**
Internships	**Coding_Score**	**Communication_Rating**	**Placement_Status**

YOUR QUESTIONS

1. What is the target?
2. Which columns are useful / excluded?
3. What data-quality issues would you inspect?
4. What preprocessing is needed?
5. Regression or classification?
6. Which model family first?
7. How would you split the data?
8. Which metrics?
9. Train 99%, Test 58% — what happened?
10. What would you change next?